STD: Generating Realistic and Diverse Safety-Critical Scenarios for Autonomous Vehicles

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***Abstract*—Autonomous driving technology has experienced significant advancements over the past decade. Developing autonomous vehicles requires extensive testing with realistic and challenging safety-critical scenarios, such as collisions, to evaluate and improve the performance of driving planners. However, due to the long-tail effect, sparsity, and rare occurrence of such scenarios in real-world datasets, obtaining sufficient data remains challenging. To address this, we propose Scenario Transformer Diffusion (STD), a methodology for the automated generation of diverse and realistic safety-critical scenarios alongside expert trajectories. To ensure realism, we designed a diffusion-based traffic generation model that enhances agent interactions, captures temporal dynamics, and reduces reliance on original trajectories. To reduce the sparsity of safety-critical scenarios, we introduced a safety-critical scenario generation module that incorporates adversarial, stability, and realism factors to guide adversarial agents toward collision scenarios. Additionally, our expert trajectory optimization module identifies solutions within these generated scenarios, providing valuable data for enhancing data-driven planners. Extensive experiments demonstrate that STD advances the state-of-the-art in generating stable, realistic, and diverse safety-critical scenarios. Furthermore, STD not only effectively tests planner performance but also enhances the training data distribution through its expert-generated trajectories, ultimately contributing to improved planner robustness and safety in real-world applications.**

***Index Terms*—Autonomous Vehicle, Scenario Generation, Diffusion Model**

I. Introduction

In recent years, with the rapid development of autonomous driving technology, data-driven algorithms have been extensively employed in the domains of autonomous driving planning and control [1]–[3]. These algorithms, leveraging large amounts of real-world driving data, are capable of learning to handle complex and dynamic traffic scenarios, demonstrating remarkable adaptability. Compared to traditional rule-based algorithms, data-driven models can flexibly respond to diverse driving situations, establishing themselves as the foundational technology in contemporary autonomous driving systems [4].

The safety of autonomous driving is a critical factor in assessing the technological maturity of this field [5], [6]. In real-world scenarios, autonomous driving systems must handle a wide range of complex situations, particularly in sudden and extreme conditions, to ensure vehicle safety. A major challenge in achieving this level of safety is the scarcity of low-frequency, high-risk safety-critical scenarios, which occur roughly once every 6.1 million miles of driving [7], [8]. For training purposes, autonomous driving systems may lack sufficient training data to handle safety-critical scenarios, which can lead to incorrect decisions in these situations and ultimately cause the system to fail. For testing, both data-driven and traditional rule-based autonomous driving planners require evaluation within safety-critical scenarios to verify their robustness [9], [10]. However, this type of data exhibits a "long-tail effect" [11], [12]—while large volumes of data have been collected to cover typical everyday driving situations, safety-critical scenarios remain exceedingly rare. Therefore, obtaining a diverse collection of low-frequency, high-risk safety-critical scenarios is essential.

We believe that an ideal set of safety-critical scenarios should satisfy two key criteria: (1) the scenarios should be comprehensive, realistic, and diverse, as previously mentioned, to effectively overcome the long-tail effect, and (2) they should be tested and utilized in an end-to-end manner to effectively evaluate the actual performance of planners under low-frequency, high-risk conditions. End-to-end testing is crucial, as it offers a direct assessment of the scenarios' quality and effectiveness, ensuring they meet the practical requirements and expectations of users, rather than merely being evaluated through metrics [13], [14].

Currently, there are two primary methods used to generate these scenarios: sampling from real-world data and scenario generation in simulated environments [15].

The first method involves sampling safety-critical scenarios from real-world driving data. [16]–[20] ensure high realism in the generated scenarios. However, safety-critical scenarios are extremely rare in everyday driving data and do not cover all potential extreme situations, making this method inefficient and limited.

A natural alternative method is to automatically generate safety-critical scenarios in simulated environments, enabling the efficient generation of diverse scenarios. Although substantial progress has been made in recent studies, there are still

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three major limitations. These limitations impact the realism and diversity of the generated scenarios and hinder their effectiveness in the actual testing and performance improvement of autonomous driving planners.

L1: Neglecting dynamic interactions in complex traffic environments results in unrealistic and impractical scenario generation. For instance, methods such as [21]–[25] optimize pre-selected adversary trajectories to collide with the ego agent in environments with a limited number of agents. However, in complex traffic settings, adversarial agents may collide prematurely with other non-target agents, making it difficult to generate the intended adversarial scenarios.

L2: Dependence on original trajectories limits scenario diversity. For instance, methods like [26], [27] impose adversarial constraints on original trajectories to trigger collisions and generate target scenarios. While these methods ensure realism, they exhibit certain limitations when it comes to generating more diverse safety-critical scenarios within the same environment. This falls short of the previously mentioned criteria of generating both realistic and diverse scenarios. Furthermore, these methods require the presence of original trajectories to function effectively.

L3: Lack of practical application of generated scenarios for testing or improving planner performance. For example, [21], [28]–[30] do not adequately evaluate the actual impact of these scenarios on planner performance, a factor of primary concern to users. Moreover, these studies do not confirm their effectiveness in improving planner performance, limiting the practical value of these methods for optimizing real-world autonomous driving systems.

To address these three limitations, we propose Scenario Transformer Diffusion (STD), a query-centric multi-agent traffic generation model for safety-critical scenarios. STD integrates adversarial guidance and data-driven generation, improving dynamic interactions, enhancing scenario realism and diversity, and boosting the performance of autonomous driving planners. Below, we detail how STD addresses these three limitations.

M1: Improvements in dynamic interactions within complex traffic environments. We designed the Social Transformer and Decoder components to enhance agent interaction and temporal comprehension. For guidance, our safety-critical scenario generation module includes collision guidance, reducing unintended collisions with non-target agents. Both measures improve the generation of safety-critical scenarios in complex traffic environments.

M2: Overcoming dependence on original trajectories for scenario generation. STD generates future agent trajectories using environmental conditions (initial states, attributes, and maps) without relying on original trajectories. This allows us to generate more diverse safety-critical scenarios, improving coverage of sparse cases.

M3: Practical application of safety-critical scenario generation for enhancing and testing planners. STD not only generates safety-critical scenarios but also provides expert trajectories to handle these scenarios. By fine-tuning planners with these expert trajectories, we effectively enhance planners' performance. Additionally, we test the performance of existing planners in safety-critical scenarios for a more comprehensive evaluation.

Through these measures, STD effectively overcomes existing limitations, achieving notable advancements in generating safety-critical scenarios and optimizing planner performance. In summary, our contributions are as follows:

- We propose STD, a query-centric multi-agent traffic generation model designed specifically to generate safety-critical scenarios.
- We propose a safety-critical scenario generation module and an expert trajectory optimization module to enhance the STD's generation process. The first module focuses on generating realistic and feasible safety-critical scenarios, while the second provides expert solutions tailored to these scenarios, effectively integrating scenario generation with adaptive response strategies.
- We leverage these expert solutions to fine-tune autonomous driving planners, significantly enhancing their performance and robustness in handling safety-critical scenarios.
- We evaluate and demonstrate the performance of the generated safety-critical scenarios and their corresponding expert trajectories, conducting a clustering analysis on the types of collisions within these scenarios. Additionally, we experimentally demonstrate the effectiveness of the generated safety-critical scenarios in testing and enhancing the performance of autonomous driving planners.

II. Preliminaries

A. Problem Definition

We consider an interactive traffic environment where the ego vehicle and surrounding vehicles are represented as $\mathcal{V} = \{V_1, \cdots, V_N\}$, driving on a complex multi-lane road. The ego agent can sense the surrounding agents' states (e.g., position, heading, speed) and make decisions at each time step t over the target time horizon T.

Action. The actions of the agents set $\mathcal{V}$ at time t are represented by $m^t = [m_1^t, \cdots, m_N^t]$, where the action $m = [a, \omega]^T$ includes acceleration a and angular velocity ω.

State. At time step t, the state of the agent set $\mathcal{V}$ is $s^t = [s_1^t, \cdots, s_N^t]$, where each state $s = [x, y, \theta, \nu]^T$ includes 2D coordinates, heading, and speed. The next state s^{t+1} is computed from the current state s^t and action m^t using the dynamics model f, as $s^{t+1} = f(s^t, m^t)$.

Attribute. Each agent $\mathcal{V}$ also has attributes $Q = [l, w]$, where l and w represent the length and width, respectively.

Vector Map. The local vector map of the agent set $\mathcal{V}$ at time step t is represented as $\mathcal{I}^t \in \mathbb{R}^{N \times Z \times P \times R}$, where Z is the number of polylines in the map, P is the number of points on each polyline, and R is the number of attributes per point (position and angle).

Decision-making Context. The decision-making context of the agents is represented as $C = \left\{\mathcal{S}^{t-T_h:t}, Q, \mathcal{I}^t\right\}$, including the historical states $\mathcal{S}^{t-T_h:t} = \left\{s^{t-T_h}, \cdots, s^t\right\}$ over the past T_h time steps, their attributes Q, and the local vector map $\mathcal{I}^t$.

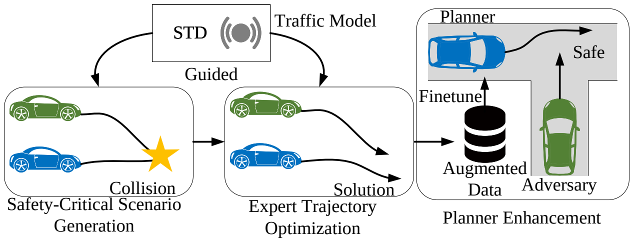


Fig. 1. Framework Overview

Time Step. We discretize the continuous time horizon into time steps, with a total length of $T_h + T$, where T_h is the historical length and T is the length to be generated. The time interval between consecutive steps is Δt.
Trajectory. The generated trajectory $\tau = [\tau_a, \tau_s]^T$ consists of the action sequence $\tau_a = \left[m^0, \cdots, m^{T-1}\right]^T$ and the state sequence $\tau_s = \left[s^1, \cdots, s^T\right]^T$. STD generates only the action trajectory τ_a, and to ensure physical feasibility, τ_s is derived from τ_a: $\tau_s = f(s^0, \tau_a)$.
Objective. Our objective is twofold: first, to select an appropriate surrounding agent as an adversary and generate a trajectory that leads to a collision with the ego agent, thus forming a safety-critical scenario. Second, to generate a feasible trajectory for the ego agent in this scenario to avoid the collision, which we call the expert trajectory.

B. Conditional Diffusion Probabilistic Models

Diffusion probabilistic models are deep generative models that have achieved state-of-the-art results in fields such as image and video generation [31]–[34]. They generate samples consistent with the original data distribution by adding noise to the samples and learning the reverse denoising process. Diffusion probabilistic models can be formalized as two Markov chain processes of length K, referred to as the diffusion process and the reverse process. Conditional diffusion probabilistic models are similar to unconditional diffusion probabilistic models but consider conditional information C in each step of the diffusion process. Let $\tau^0 \sim p$, where p is the probability distribution of the model trajectory, and τ^k is the sampled latent variable trajectory, where $k = 1, \cdots, K$ are the diffusion steps. $\tau^K \sim \mathcal{N}(0, I)$, where $\mathcal{N}$ denotes the Gaussian distribution. The diffusion process gradually adds Gaussian noise to τ^0 until τ^0 approaches τ^K, while the reverse process denoises τ^K considering the conditional information C to recover τ^0.

III. Methodology

A. Framework Overview

Figure 1 illustrates the overall framework of the proposed STD, comprising four main components: traffic model, safety-critical scenario generation module, expert trajectory optimization module, and planner enhancement. STD generates safety-critical scenarios along with expert trajectories that address these scenarios, thus providing more challenging training data for autonomous driving planners. Additionally, the generated safety-critical scenarios are used to evaluate the performance of autonomous driving planners under extreme conditions.
Traffic Model is designed to generate stable, realistic, and diverse traffic scenarios. Its inputs consist of trajectory noise and initial conditions, including agents' states, attributes, and map. Based on these conditions, the model incrementally denoises the trajectory noise through diffusion to produce high quality traffic scenarios.
Safety-Critical Scenario Generation Module guides the adversarial agent to minimize its distance from the ego agent at each diffusion denoising step to create collision scenarios. In this module, we designed a hybrid guidance function composed of adversarial, environmental, collision, and initialization guidance, achieving highly effective results in the generated scenarios. Specifically, adversarial guidance increases the likelihood of collisions between the adversarial and ego agents, while other guidance ensure that the generated scenarios are both usable and realistic.
Expert Trajectory Optimization Module assists traffic model in identifying expert solutions, or expert trajectories, within generated safety-critical scenarios. This module also employs a hybrid guidance function, which includes original, environmental, collision, and initialization guidance, to regenerate the ego agent's trajectory as a stable, feasible, and realistic expert trajectory.
Planner Enhancement. We design a workflow to evaluate the real performance of the generated safety-critical scenarios and expert trajectories. Specifically, we use expert trajectories as training data to fine-tune pre-trained planners, thereby effectively enhancing the fine-tune planners' ability to handle safety-critical scenarios.

B. Scenario Transformer Diffusion Model

In our approach, trajectories are generated through an iterative denoising process, which is learned by reversing a predefined diffusion process. As stated in Section II, the state sequence τ^s is derived from the action sequence τ^a through the dynamics model f, expressed as $\tau^s = f(s^0, \tau^a)$. The trajectory input to the model is $\tau = [\tau^a, \tau^s]^T$. Starting with a clean trajectory sampled from the data distribution, $\tau^0 \sim q(\tau^0)$, the forward noising process introduces Gaussian noise at each step k, generating a sequence of trajectories with progressively increasing noise $(\tau^0, \cdots, \tau^K)$:

$$q(\tau^{1:K}|\tau^0) := \prod_{k=1}^{K} q(\tau^k|\tau^{k-1}) \tag{1}$$

$$q(\tau^k|\tau^{k-1}) := \mathcal{N}(\tau^k; \sqrt{1-\beta_k}\tau^{k-1}, \beta_k I) \tag{2}$$

where β_k is the variance at each step, and with sufficiently large K, we approximate $q(\tau^K) \approx \mathcal{N}(\tau^K; 0, I)$.

STD learns this reverse process, allowing noisy samples to be denoised back into reasonable trajectories. Each step of this reverse process is conditioned on the agent's decision-making context C:

$$p_\theta(\tau^{0:K}|C) := p_\theta(\tau^K) \prod_{k=1}^{K} p_\theta(\tau^{k-1}|\tau^k, C) \tag{3}$$

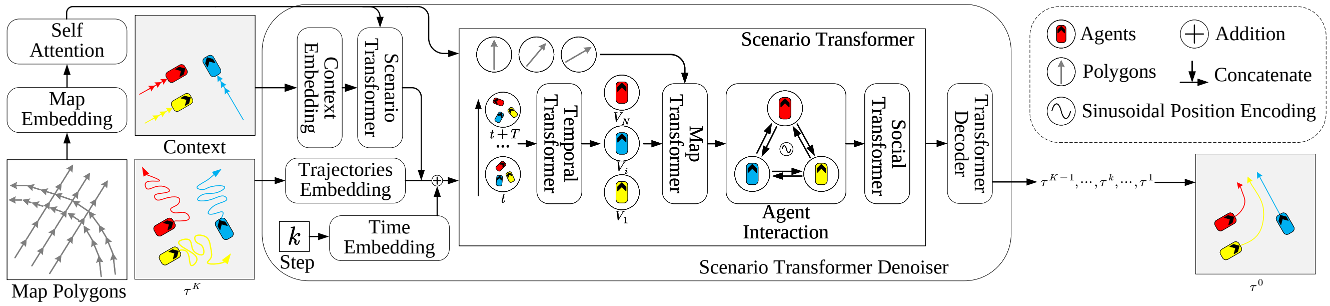


Fig. 2. Detailed Architecture of STD Denoiser

$$p_\theta(\tau^{k-1}|\tau^k, C) := \mathcal{N}(\tau^{k-1}; \mu_\theta(\tau^k, k, C), \Sigma_\theta(\tau^k, k, C)) \quad (4)$$

where θ are the model parameters. According to [31], [32], the Gaussian transition variance is fixed as $\Sigma_\theta(\tau^k, k, C) = \Sigma_k = \sigma_k^2 I = \beta_k I$.

The architecture of STD is depicted in Figure 2. Initially, we embed the agent's decision-making context and map its feature dimensions to d dimensions, processed through L_c stacks of the Scenario Transformer, resulting in the transformed features $F'_C \in \mathbb{R}^{T_h \times d}$. Additionally, we process the map polygons by embedding them into d dimensions, denoted as $F_M \in \mathbb{R}^{L \times d}$. We utilize a multi-head self-attention mechanism to extract features among the map polygons, helping the agent effectively leverage localized vectorized maps. The self-attention mechanism for map feature encoding can be represented as:

$$F'_M[i] = \mathrm{MHA}\begin{pmatrix} \mathrm{Q}: [F_M[i] + \mathrm{PE}] \\ \mathrm{K}: \{[F_M[j] + \mathrm{PE}]\}_{j \in \Omega(i)} \\ \mathrm{v}: \{[F_M[j] + \mathrm{PE}]\}_{j \in \Omega(i)} \end{pmatrix} \quad (5)$$

where $i \in \{1, \cdots, L\}$, $\Omega(i)$ is the set of map features i needs to process. $\mathrm{MHA}(\cdot)$ denotes the multi-head attention mechanism. PE represents the sinusoidal encoding [35], [36] for processing the j-th map feature. $F'_M[i] \in \mathbb{R}^d$ is the output feature of the map self-attention mechanism.

We encode the noisy trajectory as $F_\tau \in \mathbb{R}^{T \times d}$ and concatenate it with the agent's decision-making context. Additionally, we combine the result with the encoded denoising step features $F_t \in \mathbb{R}^{(T_h + T) \times d}$, represented as:

$$F_A = \mathrm{Concat}(F_\tau, F'_C) + F_t \quad (6)$$

where $\mathrm{Concat}(\cdot)$ denotes the concatenation operation. The resulting feature $F_A \in \mathbb{R}^{(T_h+T)\times d}$ is input to L_{enc} layers of stacked Motion Transformer, outputting $F'''_A \in \mathbb{R}^{(T_h+T)\times d}$. Finally, through a standard Transformer decoder followed by a three-layer MLP, the denoised trajectory τ^{k-1} is obtained. Specifically, our Transformer decoder's multi-head cross-attention mechanism can be represented as:

$$F_{out}[t] = \mathrm{MHA}\begin{pmatrix} \mathrm{Q}: [F'''_A[t] + \mathrm{PE}] \\ \mathrm{K}: \{[F'''_A[\iota] + \mathrm{PE}]\}_{\iota \in \Omega(t)} \\ \mathrm{V}: \{[F'''_A[\iota] + \mathrm{PE}]\}_{\iota \in \Omega(t)} \end{pmatrix} \quad (7)$$

where $t \in \{T_h + 1, \cdots, T_h + T\}$, $\Omega(t)$ is the set of time features that need to be processed at time step t, with the number of elements in $\Omega(t)$ being $T_h + T$. PE represents the sinusoidal encoding for processing the t-th time feature. $F_{out} \in \mathbb{R}^{T \times d}$ is the output of the Transformer decoder, and finally, we output the denoised trajectory τ^{k-1} through a three-layer MLP:

$$\tau^{k-1} = f(s^0, \mathrm{MLP}(F_{out})) \quad (8)$$

Scenario Transformer. Figure 2 also shows the structure of the Scenario Transformer, which is composed of three parts: Temporal, Map, and Social Transformer. Our network architecture is entirely Transformer-based, with two key modifications: (1) Unlike [37], [38] that rely on a global coordinate system centered on a focal agent, we use a query-centric approach. This models agent relationships symmetrically, decoupling from any global system and reducing redundant computation, enabling parallel multi-agent traffic generation. (2) Inspired by [39], [40], we use a position encoder to apply sinusoidal encoding to relative agent relationships, ensuring consistent position similarity.

Temporal Transformer analyzes encoded trajectories to effectively capture and understand dynamic interactions and relationships among agents over time. This process provides crucial temporal information for scenario generation, enabling the system to accurately model and respond to changes in complex environments. Specifically, the self-attention mechanism of the Temporal Transformer can be represented as:

$$F'_A[t] = \mathrm{MHA}\begin{pmatrix} \mathrm{Q}: [F_A[t] + \mathrm{PE}] \\ \mathrm{K}: \{[F_A[\iota] + \mathrm{PE}]\}_{\iota \in \Omega(t)} \\ \mathrm{V}: \{[F_A[\iota] + \mathrm{PE}]\}_{\iota \in \Omega(t)} \end{pmatrix} \quad (9)$$

where $t \in \{1, \cdots, T_h + T\}$, $\Omega(t)$ is the set of token that need processing at time step t. $F'_A[t] \in \mathbb{R}^d$ is the output of the Temporal Transformer.

Map Transformer employs a cross multi-head attention mechanism, using the central agent's features as queries and local vectorized map features as keys and values. Map Transformer aims to enable agents to capture surrounding map environments to ensure driving within feasible areas. Specifically, the cross-attention mechanism of the Map Transformer can be represented as:

$$F''_A[i] = \mathrm{MHA}\begin{pmatrix} \mathrm{Q}: F'_A[i] \\ \mathrm{K}: \{[F'_M[j] + \mathrm{PE}]\}_{j \in \Omega(i)} \\ \mathrm{V}: \{[F'_M[j] + \mathrm{PE}]\}_{j \in \Omega(i)} \end{pmatrix} \quad (10)$$

where $i \in \{1, \cdots, N\}$, $\Omega(i)$ is the set of token that need processing at agent V_i. $F_A^{''}[i] \in \mathbb{R}^d$ is the output of the Map Transformer.

Social Transformer overcomes the limitations of agent-centric coordinate systems [41] by employing a query-centric self-attention mechanism to capture relative agent relationships. This leverages the symmetry and repeatability of interactions, reducing redundant calculations and enhancing performance [42], [43]. Additionally, a position encoder sinusoidally encodes these relationships [39], [40], ensuring consistent relative position similarity and avoiding unnecessary dimensional expansion.

To explore the relative relationships between the query agent and other agents in the scene, we transform the coordinate systems of other agents into the query agent's coordinate system using a transformation matrix:

$$R^{pos}[i,j] = (P[j] - P[i]) \begin{bmatrix} \cos H[i] & -\sin H[i] \\ \sin H[i] & \cos H[i] \end{bmatrix} \tag{11}$$

$$R^{\theta}[i,j] = H[j] - H[i] \tag{12}$$

$$R^{\nu}[i,j] = (\nu[j] - \nu[i]) \begin{bmatrix} \cos H[i] & -\sin H[i] \\ \sin H[i] & \cos H[i] \end{bmatrix} \tag{13}$$

$$R[i,j] = \left[R^{pos}[i,j], R^{\theta}[i,j], R^{\nu}[i,j]\right] \tag{14}$$

where $R^{pos}[i,j]$, $R^{\theta}[i,j]$, and $R^{\nu}[i,j]$ represent the 2D relative position, relative heading angle, and relative velocity between agent V_i and agent V_j, respectively.

The cross-attention mechanism of the Social Transformer can be represented as:

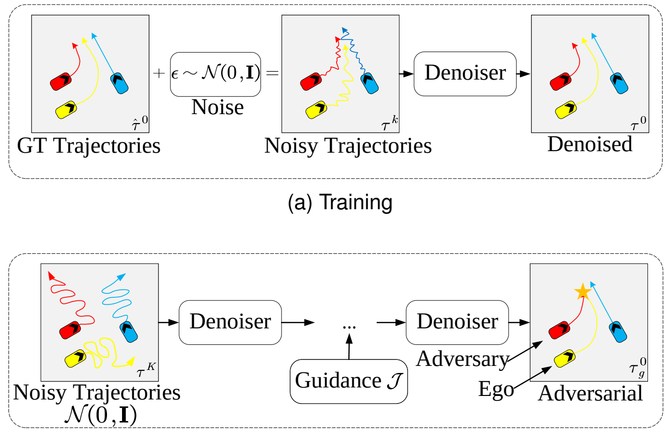
$$F_A^{'''}[i] = \mathrm{MHA}\left(\begin{array}{l} \mathrm{Q}: F_A^{''}[i] \\ \mathrm{K}: \{[F_M^{''}[j] + \mathrm{PE}(R[i,j])]\}_{j \in \Omega(i)} \\ \mathrm{V}: \{[F_M^{''}[j] + \mathrm{PE}(R[i,j])]\}_{j \in \Omega(i)} \end{array} \right) \tag{15}$$

where $\mathrm{PE}(\cdot)$ represents sinusoidal encoding using the relative position inside parentheses to ensure consistency in relative position similarity [39]. $F_A^{'''}[i] \in \mathbb{R}^d$ is the output of the Social Transformer.

After passing through the Temporal, Map, and Social Transformer, we maintain consistent input and output dimensions, enabling effective stacking of these structures for improved interactions among temporal sequences, maps, and agents in traffic scenarios.

Training. During the STD training phase, the network directly predicts the mean μ of the clean trajectory τ^0, instead of predicting the next noisy trajectory τ^{k-1}. We uniformly sample the denoising step $k = \mathcal{U}(1, K)$ from denoising steps $[1, K]$. We directly add the noise loaded at step τ^k to the real trajectory τ^0 using the formula $\tau^k = \sqrt{\bar{a}_k}\tau^0 + \sqrt{1-\bar{a}_k}\epsilon$, where $\epsilon \sim \mathcal{N}(0, \mathrm{I})$ and $\bar{a}_k = \prod_{l=0}^{k}(1-\beta_l)$. We denote the model's direct output as $\hat{\tau}^0 = D_\theta(\tau^k, k, C)$, as shown in Figure 3a. The supervised training loss is given by:

$$\mathcal{L}_{loss} = \mathbb{E}_{\epsilon,k,\tau^0,C} \left\| \tau^0 - \hat{\tau}^0 \right\|^2 \tag{16}$$



(a) Training

(b) Guided Sampling

Fig. 3. Training and Sampling

C. Safety-Critical Scenario Generation

The purpose of STD is to generate safety-critical scenarios that do not exist in the original dataset. Through these scenarios, experts (Section III-D) can identify solutions to tackle adversarial challenges and utilize these expert trajectories to improve the performance of planners. To generate safety-critical scenarios, we incorporate a specific module into STD that guides adversarial agents to intentionally collide with the ego agent.

Guided Sampling. Safety-critical scenario generation typically falls into two categories: one involves generating trajectories using traffic priors and then guiding them with a guidance function [26]; the other directly uses trajectories from datasets, guided by a guidance function [21], [22]. STD is distinct in that it incorporates a guidance function into each denoising step, ensuring both the stability and authenticity of the generated safety-critical scenarios. Additionally, through multiple sampling, the inherent randomness of noise allows STD to generate a diverse set of safety-critical scenarios. A schematic of the guided sampling is shown in Figure 3b.

Inspired by [44]–[46], we reformulate the reverse process as:

$$p_\theta\left(\tau_g^{k-1} | \tau_g^k, C\right) := \mathcal{N}\left(\tau_g^{k-1}; \mu_\theta + \Sigma_\theta \nabla \mathcal{J}(\mu_\theta), \Sigma_\theta\right) \tag{17}$$

where $\mathcal{J}$ is referred to as the guidance function. τ_g^k denotes the denoised trajectory generated after guided sampling at step k.

In practice, when generating trajectories, we guide sampling from multiple samples produced by STD and, at the end of the denoising process, select the final output trajectory based on the minimum value of the guidance function $\mathcal{J}$. This method ensures that the generated trajectories more closely align with the guidance requirements, thereby enhancing the quality and reliability of the sample trajectories.

Algorithm 1 implements this guided sampling process within a closed-loop simulation framework.

Safety-Critical Scenario Generation Module. Our goal is to create situations where an adversarial agent collides with the ego agent. To ensure the rationality and effectiveness of safety-critical scenarios, we do not choose adversaries that are always

Algorithm 1: STD-Guided Closed-Loop Simulation

Input: STD model μ_θ, Dynamics model f, Simulation environment $\mathcal{E}$, Denoising steps K, Initial state s^0, Historical trajectory τ_h, Guidance function $\mathcal{J}$, Guidance iterations N, Covariance matrices Σ_k, Scaling factor α, Decision frequency F, Simulation duration T

Output: Simulated trajectory τ_{sim}

$\tau_{\text{sim}} \leftarrow \emptyset$;
$s \leftarrow s^0$;
$\Delta t \leftarrow 1/F$;
for $t = 0, \Delta t, \cdots, T$ **do**
  $\tau_h \leftarrow \tau_h[\Delta t :] \oplus \tau_{\text{sim}}[t - \Delta t : t]$;
  $C \leftarrow \mathcal{E}(\tau_h)$;
  $\tau_a^K \sim \mathcal{N}(0, I)$;
  $\tau_s^K \leftarrow f(s; \tau_a^K)$;
  $\tau^K \leftarrow [\tau_a^K; \tau_s^K]$;
  for $k = K, K-1, \cdots, 1$ **do**
    $\tau_g^k \leftarrow \mu_\theta(\tau^k, k, C)$;
    for $n = 1, 2, \cdots, N$ **do**
      $\tau_g^k \leftarrow \tau_g^k + \alpha \cdot \Sigma_\theta \nabla_\tau \mathcal{J}(\tau_g^k)$;
    end
    $\tau^{k-1} \sim \mathcal{N}(\tau_g^k, \Sigma_k)$;
    $\tau_s^{k-1} \leftarrow f(s; \tau_a^{k-1})$;
    $\tau^{k-1} \leftarrow [\tau_a^{k-1}; \tau_s^{k-1}]$;
  end
  $a_t \leftarrow \tau_a^0[0]$;
  $s_{\text{next}} \leftarrow f(s, a_t)$;
  $\tau_{\text{sim}} \leftarrow \tau_{\text{sim}} \cup \tau^0[0 : \Delta t]$;
  $s \leftarrow s_{\text{next}}$;
end
return τ_{sim}

behind the ego agent. Our criterion for selecting adversaries is defined as:

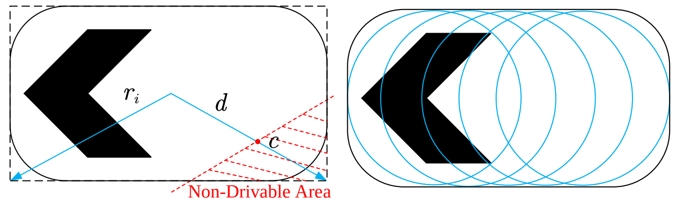
$$(i^*, t^*) = \arg\min_{i,t} \|P_i^t - P_1^t\| \tag{18}$$

where the ego agent is denoted as V_1. P_i^t represents the global coordinates of agent V_i at time step t, and t^* is the time step when the adversarial agent V_{i^*} is closest.

In the safety-critical scenario generation module, we designed an innovative hybrid guidance function composed of adversarial, environmental, collision and initialization guidance. Thus, our optimization objective is:

$$\min_{\mu} (\mathcal{J}_{adv} + \mathcal{J}_{env} + \mathcal{J}_{coll} + \mathcal{J}_{init}) \tag{19}$$

Each guidance function has a corresponding weight, denoted as w_{adv}, w_{env}, w_{coll}, and w_{init}, and we use the Adam optimizer to optimize the minimum objective function o times.
Adversarial Guidance aims to make the selected adversary more likely to collide with the ego agent by minimizing



(a) (b)

Fig. 4. Environmental Guidance (a) and Collision Guidance (b) Schematics

the distance at each time step t in two-dimensional position. Specifically, the adversarial guidance is defined as:

$$\mathcal{J}_{adv} = \sum_t \xi^t \|P_{i^*}^t - P_0^t\|^2 \tag{20}$$

$$\xi^t = \frac{\exp(-\|P_{i^*}^t - P_0^t\|)}{\sum_t \exp(-\|P_{i^*}^t - P_0^t\|)} \tag{21}$$

where ξ^t is defined as the softmin of the 2D distance between V_{i^*} and V_1 at time step t.
Environmental Guidance penalizes agents for driving in non-drivable areas. It detects this by checking the overlap between gridded non-drivable map layers and agent boundaries as shown in Figure 4a [47]. Specifically, the environmental loss can be represented as:

$$\mathcal{L}_{env}(i) = \begin{cases} 1 - d/r_i & \text{if overlap} \\ 0 & \text{otherwise} \end{cases} \tag{22}$$

where d is the distance between the collision point c and the agent center, and r_i is half of the diagonal of agent V_i's bounding box.

The environmental guidance is the sum of the environmental losses of all agents under time decay:

$$\mathcal{J}_{env} = \frac{1}{N} \sum_i \sum_t \gamma^t \mathcal{L}_{env}(i) \tag{23}$$

where γ is a decay factor.
Collision Guidance is to ensure that only the adversary and the ego agent collide in the generated safety-critical scenarios, avoiding collisions between other agents. Specifically, we use pairwise collision loss and efficient differentiable relaxation to simplify optimization. We approximate each agent with five circles as shown in Figure 4b and calculate the L2 distance between the nearest centers of each pair of agents. It can be represented as:

$$\mathcal{L}_{coll}(i,j,t) = \begin{cases} 1 - \frac{\|P_i^t - P_j^t\|}{r_i + r_j} & \|P_i^t - P_j^t\| \leqslant r_i + r_j \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

where r_i represents the radius of agent V_i's approximating circle.

The collision guidance is the sum of collision losses of all agents under time decay:

$$\mathcal{J}_{coll} = \frac{1}{N^2} \sum_{(i,j), i \neq j \neq 1} \sum_t \gamma^t \mathcal{L}_{coll}(i,j,t) \tag{25}$$

To not compute the collision guidance between the adversary and the ego agent, we mask the corresponding values in experiments.

Initialization Guidance aims to minimize the deviation between the guided trajectory and the original trajectory. This setup ensures that the naturalness and rationality of the original trajectory are preserved to the greatest extent during the generation of safety-critical scenario. It is represented as:

$$\mathcal{J}_{init} = \frac{1}{N}\sum_{i}\sum_{t}\|P_i^t - P_{i_{init}}^t\|^2 \tag{26}$$

where $P_{i_{init}}^t$ is the global coordinate of agent V_i at the initial state.

D. Expert Trajectory Optimization

The expert trajectory optimization module aims to generate optimal expert trajectories for the ego agent in collision scenarios, enhancing its performance. In contrast, the safety-critical scenario generation module modifies adversarial agents' behavior to force collisions with the ego agent. The key distinction is that the former focuses on optimizing the ego agent's trajectory, while the latter targets the adversarial agents' trajectories. Specifically, the expert trajectory optimization guides the STD to generate ego agent trajectories, while other agents follow their original paths from the safety-critical scenarios. This approach ensures that the ego agent's trajectory is optimized for handling such scenarios, improving its ability to manage potential collisions.

In the expert trajectory optimization module, we propose a hybrid guidance function that combines original, collision, environmental and initialization guidance. The optimization objective is:

$$\min_{\mu}\left(\mathcal{J}_{ori} + \mathcal{J}_{ego} + \mathcal{J}_{env} + \mathcal{J}_{init}\right) \tag{27}$$

Similarly, each guidance function has corresponding weights, denoted as w_{ori}, w_{ego}, w_{env}, and w_{init}, and we use the Adam optimizer to optimize the minimum objective function o times.

Ego Guidance is to ensure that the ego agent does not collide with any other agents. The definition of $\mathcal{J}_{ego}$ is similar to $\mathcal{J}_{coll}$ from Section III-C, but $\mathcal{J}_{ego}$ calculates only the collision loss between the ego agent and other agents, not among all agents, and it carries a greater weight. Specifically, this can be expressed as:

$$\mathcal{J}_{ego} = \frac{1}{N}\sum_{(i,j),i=1,i\neq j}\sum_{t}\gamma^t\mathcal{L}_{coll}(i,j,t) \tag{28}$$

Original Guidance directs non-ego agents to follow the trajectories from the original safety-critical scenarios. It is defined as:

$$\mathcal{J}_{ori} = \frac{1}{N}\sum_{i,i\neq 1}\sum_{t}\gamma^t\|P_i^t - P_{i_{adv}}^t\|^2 \tag{29}$$

where $P_{i_{adv}}^t$ is the global coordinate of V_i in safety-critical scenario. Similarly, to not compute original guidance for the ego agent, we use masking in experiments to shield the corresponding values. $\mathcal{J}_{env}$ and $\mathcal{J}_{init}$ are similar to their counterparts in Section III-C but are only directed towards the ego agent, while other agents are guided by $\mathcal{J}_{ori}$.

E. Planner Enhancement

The ultimate goal of generating safety-critical scenarios is to enhance the planner's ability to respond in such scenarios [48]. To achieve this, we employed the following strategies: (1) Using STD to generate a large number of safety-critical scenarios. (2) Generating expert solutions to address these safety-critical scenarios. (3) Utilizing these solutions to fine-tune the original planner, thereby enhancing its safety and robustness in safety-critical scenarios.

To demonstrate the effectiveness of the STD-generated safety-critical scenarios and the performance of the expert solution trajectories, we designed a simple yet effective planner. Specifically, we utilize existing trajectory prediction algorithms (e.g., [49]–[51]) to generate future trajectories for the ego agent, and then employ a dynamics model [52] to translate these trajectories into actions to drive the motion of the ego agent. Through this approach, we are able to assess the planner's ability and effectiveness in handling safety-critical scenarios after fine-tuning.

IV. Experiments

A. Experimental Settings

Datasets. nuScenes [53] is a widely used dataset in autonomous driving research, covering diverse scenes and dense traffic conditions, providing 5.5 hours of manually annotated trajectories. The dataset contains 1,000 scenes, each consisting of 20-second traffic segments annotated at 2 Hz. We interpolated the nuScenes dataset to increase its sampling frequency to 10 Hz and divided it into training and validation sets. All models were trained on the training set and evaluated on the validation set. Our primary focus is on simulating moving agents as they are most relevant to control.

Simulation Environment. Our simulation environment is initialized based on the positions and historical states of agents in real driving data. Specifically, all scenes are from the validation set, ensuring that these scenes have not been seen by the trained models. The simulation runs at a frequency of 10 Hz. In practical experiments, collisions between agents in safety-critical scenarios typically occur within 10 seconds, so we set the simulation timestep to 10 seconds. To comprehensively evaluate the diversity of generated scenarios and the long-term performance of the model, the traffic simulation duration is set to 20 seconds. All experiments are conducted in a closed-loop simulation environment, with a decision frequency of 2 Hz.

Implementation Details. All experiments were conducted on an Intel® Core™ i7-13700KF CPU and a single NVIDIA® GeForce RTX™ 4090 GPU. The STD model is trained on the nuScenes training set to generate 5.2 seconds of future motion conditioned on 3 seconds of historical motion. Both motion sequences are sampled at 10 Hz, with the historical trajectory explicitly including the current state. The model was trained for 20 epochs using a batch size of 16, requiring approximately 33 hours of training time. During inference, the decision denoising process averages 11.5 seconds, while generating 10 seconds of simulation requires an average of 135.9 seconds. The specific parameters are as follows: the timestep for generating future motion is $T = 52$, the timestep for historical motion is $T_h = 31$, and the diffusion steps are $K = 100$. Regarding the details of the STD structure, the number of stacked layers of the Motion Transformer mentioned in Section III-B is $L_c = L_{enc} = 2$, and the feature dimension is $d = 128$. The local vector map covers a 50-meter radius centered on each agent, with $Z = 15$ polylines and $P = 80$ points per polyline, and the attribute dimension is $R = 3$ (including 2D coordinates and polyline orientation). The dynamics model f adopts the unicycle model [54]. During training, we consider the $N = 20$ nearest agents to the sampled agent, while during inference, the number of nearby agents is not limited. We use the AdamW [55] optimizer for training with a learning rate of 0.0001. Through sensitivity analysis and empirical parameter tuning, the weights for each guidance function in generating safety-critical scenarios are set as: $w_{\text{adv}}, w_{\text{env}}, w_{\text{coll}}, w_{\text{init}} = 1.2, 0.8, 1.2, 1.0$, while for generating expert trajectories, the weights are set as: $w_{\text{ori}}, w_{\text{ego}}, w_{\text{env}}, w_{\text{init}} = 1.0, 5.0, 0.8, 1.0$.

Baselines. Due to the different focuses of previous related works, they are not directly comparable. For example, [5], [56] focus only on generating specific scenarios, while [57], [58] attack the entire autonomous driving stack during safety-critical scenario generation, rather than just the planning module. Additionally, [59] primarily focus on generating videos of adversarial scenarios. Therefore, we selected several state-of-the-art safety-critical scenario generation methods for comparison:

- CTG++(Adv) [45]. A diffusion-based method for generating safety-critical scenarios.
- STRIVE [26]. A graph-based CVAE incorporating traffic priors for generating safety-critical scenarios.
- AdvDiffuser [60]. Guided diffusion framework for adversarial safety-critical scenarios.
- LCSim [61]. Large-scale controllable traffic simulator via guided diffusion.

Additionally, we conducted evaluations on traffic simulation performance, aiming to assess the diversity of the generated scenarios and the long-term performance of the model. Therefore, in addition to the previously mentioned methods, we also compared our approach with the following advanced traffic simulation methods:

- SimNet [62]. A deterministic behavior cloning approach for traffic simulation.
- TrafficSim [47]. A traffic simulation method utilizing isotropic Gaussian CVAE.
- BITS [63]. A traffic simulation method based on a two-layer imitation learning framework.

B. Evaluation of Safety-Critical Scenario

Metrics. In the evaluation of safety-critical scenario generation and expert trajectory optimization, we primarily focus on the **effectiveness**, **realism**, and **stability** of the scenario.

Effectiveness is assessed through the generated collision rate and the expert resolution rate. During the safety-critical scenario generation phase, effectiveness is measured by the generated collision rate, while in the expert trajectory optimization phase, it is evaluated by the expert resolution rate.

- **Generated Collision Rate (GCR)** is determined by calculating the proportion of scenarios where the ego agent collides with the adversary agent, obtained by dividing the number of collision scenarios by the total number of scenarios.
- **Expert Resolution Rate (ERR)** measures the expert's ability to find collision-avoidance trajectories in the generated safety-critical scenarios, calculated by dividing the number of successful expert solutions by the number of collision scenarios.

Realism is evaluated by comparing the statistical driving features of the generated motion with those of real-world driving data. According to [63], this comparison is performed by calculating the Wasserstein distance between the normalized histograms of the driving features of the generated motion and those of real-world motion.

- **Realism Bias (RB)** is defined as the average Wasserstein distance of the distributions for longitudinal acceleration magnitude, lateral acceleration magnitude, and jerk.

Stability is assessed by reporting the failure rate, collision rate, and off-road rate. A critical failure is defined as an agent either colliding with another agent or driving off the road for more than 1 second.

- **Failure Rate (FR)** is the average proportion of agents experiencing critical failures in the scenario.
- **Collision Rate (CR)** is the average proportion of agents colliding with another agent.
- **Off-road Rate (OR)** represents the proportion of timesteps an agent spends outside the drivable area, averaged across all agents.

During the safety-critical scenario generation phase, we primarily report the FR, CR, and OR for the adversarial agent in relation to other agents. In the expert trajectory optimization phase, the focus shifts to the ego agent, where we evaluate its FR, CR, and OR in relation to other agents. Additionally, in metrics such as AOR, EOR, OFR, and OCR, the letters A, O, and E denote adversarial agent, other agent, and ego agent, respectively.

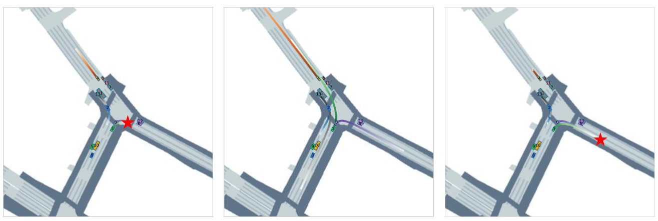
Table I presents the quantitative results for safety-critical scenario generation. We evaluated various methods using metrics such as GCR, RB, Adv OR (AOR), Other FR (OFR), and Other CR (OCR). Except for Adv OR, which is only 0.01% lower than STRIVE, STD outperforms all baseline methods across GCR, RB, OFR, and OCR, demonstrating its ability to generate stable and realistic safety-critical scenarios. This superior performance is attributed to STD being a generative model trained on real-world datasets, ensuring that output trajectories conform to real-world distributions. Additionally, the inclusion of Social Transformer and Decoder components enhances the model's capability to handle complex dynamic traffic environments (see Section IV-G for details). Our safety-

TABLE I
EVALUATION OF SAFETY-CRITICAL SCENARIO GENERATION

Method	GCR ↑	RB ↓	AOR ↓	OFR ↓	OCR ↓
STRIVE	43.00	0.088	**0.00**	9.28	8.79
CTG++(Adv)	71.00	0.190	1.31	6.73	5.92
LCsim	68.00	0.068	1.15	8.83	7.01
AdvDiffuser	70.00	0.080	0.01	5.82	5.48
STD	**73.00**	**0.060**	0.01	**5.18**	**4.65**

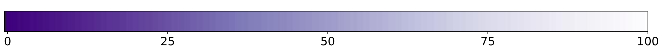
STD CTG++(Adv) STRIVE

(a)



STD CTG++(Adv) STRIVE

(b)



(c) Timestep and Trajectory Color Correspondence

Fig. 5. Comparison of STD and Baseline across Different Scenarios

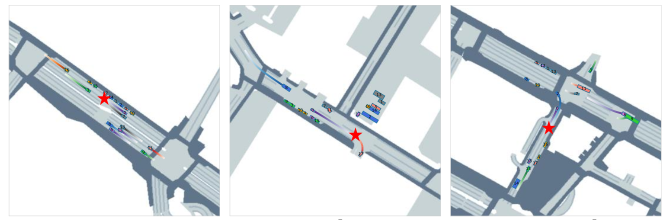
critical scenario generation module employs a hybrid guidance function, ensuring effective scenario generation while maintaining realism and stability. Although STD, as a generative model, exhibits a certain off-road rate, STRIVE mitigates this by aligning generated trajectories with real-world data. However, STRIVE's dependence on test datasets limits the diversity of its generated scenarios. Overall, these performance metrics highlight STD's efficacy in generating diverse, stable, and realistic safety-critical scenarios.

To better visualize the comparison, we demonstrate the advantages of STD through two examples using only CTG++(Adv), STRIVE, and STD. In Figure 5a, both CTG++(Adv)'s Agent 29 and STRIVE's Agent 30 approach the ego agent, compromising scenario realism. In contrast, STD avoids this issue. In Figure 5b, due to STRIVE and CTG++(Adv) optimizing all potential adversaries, STRIVE takes longer to generate safety-critical scenarios, and CTG++(Adv) even fails to find a reasonable solution. STD successfully avoids these issues. In Figure 5c, the trajectory color indicates the corresponding timesteps. Unless stated otherwise, all trajectories in this paper follow this mapping.

Table II presents the quantitative metrics for expert solutions generated in safety-critical scenarios. STD outperforms all baseline methods in the ERR, RB, Ego OR (EOR), Other FR (OCR), and Other CR (OCR) metrics. While its EOR performance surpasses CTG++(Adv), AdvDiffuser, and LCSim, it remains slightly below STRIVE. This demonstrates that STD not only generates stable and realistic safety-critical scenarios but also finds feasible and realistic expert trajectories within these scenarios. High-quality expert trajectories provide more comprehensive training data for data-driven autonomous driving planners, effectively reducing the issue of data sparsity in safety-critical scenarios.

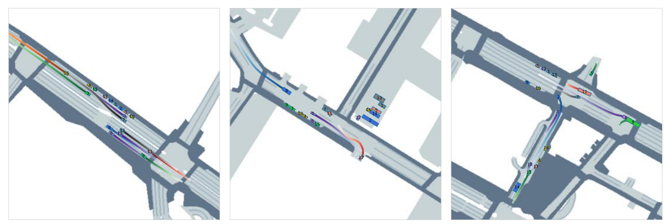
TABLE II
EVALUATION OF EXPERT TRAJECTORIES GENERATION

Method	ERR ↑	RB ↓	EOR ↓	OFR ↓	OCR ↓
STRIVE	48.84	0.075	**0.65**	11.34	11.65
CTG++(Adv)	59.15	0.234	6.54	15.26	14.08
LCSim	60.29	0.115	12.24	15.56	13.61
AdvDiffuser	66.66	0.106	13.71	14.04	12.42
STD	**75.34**	**0.060**	6.14	**8.29**	**8.17**



Cut-in Head-on Rear-end

(a) Safety-Critic Scenario



(b) Expert Solution

Fig. 6. Expert Trajectories in Safety-Critical Scenarios

In Figure 6, we present three types of cases where STD successfully found expert solutions in the generated safety-critical scenarios. Based on the statistics from [64], we selected the most common cases in daily traffic: from left to right, a cut-in collision, a head-on collision, and a rear-end collision. STD is able to generate stable and realistic expert solutions for all three types of safety-critical scenarios.

C. Evaluating Generated Scenario with Planner

Metrics. To further illustrate the effectiveness of STD, we evaluated the performance of AutoBot [49] in both adversarial scenarios generated by STD and regular scenarios.

- **Planner's collision rate (PCR)** is defined as the number of collision scenarios divided by the total number of scenarios.

We measured effectiveness by recording the planner's collision rate, as shown in the "Pre-Trained" row of Table III. In the adversarial scenarios generated by STD, the Pre-Trained AutoBot exhibited a 24.00% increase in PCR, indicating that the scenarios generated by STD effectively test the safety performance of the planner in extreme environments. Compared to the GCR during scenario generation, the planner's

TABLE III
PLANNER PERFORMANCE IN REG. AND ADV. SCENARIOS

Planner	Scenario	PCR ↓	OR ↓	AV ↑	Jerk ↓
Pre-Trained	Regular	20.00	3.54	4.77	1.05
	Adversarial	44.00	2.34	4.75	1.04
Fine-Tuned	Regular	19.00	9.70	6.04	0.65
	Adversarial	39.00	8.85	6.20	0.62

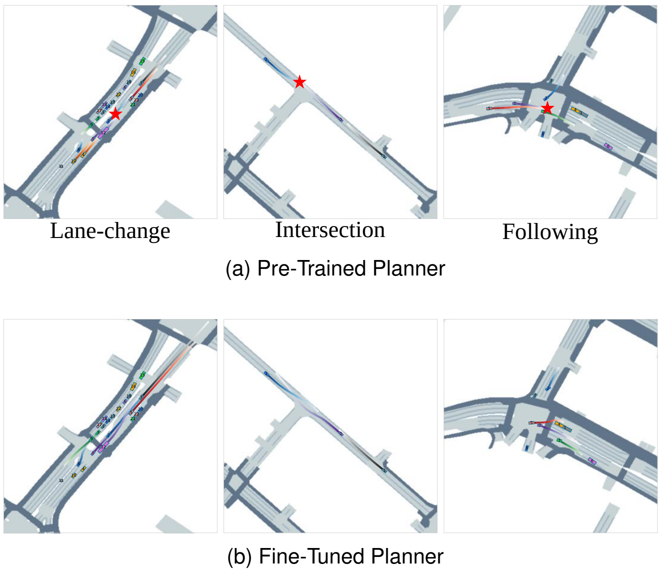


Fig. 7. Different Planners in Safety-Critical Scenarios

PCR in adversarial scenarios was significantly lower. However, the metrics during scenario generation only reflect offline evaluations, while the actual impact of the generated scenarios when applied to the planner is what truly matters to the user. We conducted a comprehensive evaluation of the planner's actual performance in generated scenarios, an aspect that has not been thoroughly explored in other works such as [21], [28], [45], [65]. Our results, focusing on real-world application, provide a more accurate reflection of how generated scenarios influence the planner's performance, bridging the gap left by purely offline evaluation studies.

D. Evaluation of Planner Enhancement

In this subsection, we analyze the effectiveness of the generated expert trajectories in enhancing the original training data distribution. Specifically, we evaluate the performance of the autonomous driving planner, AutoBot, before and after fine-tuning with expert trajectories, to assess the effectiveness of the generated safety-critical scenarios and their corresponding expert trajectories.

Metrics. We measure the planner's performance by evaluating the PCR, OR, average velocity, and jerk. The PCR and OR have already been defined in Section IV-C.

• **Average Velocity (AV)** is used to assess the efficiency of the autonomous vehicle [66], [67]. A higher AV indicates better efficiency.

• **Jerk**, defined as the rate of change of acceleration, measures driving comfort due to its significant impact on passenger experience [68]. Higher jerk values indicate greater passenger discomfort. We assess the planner's comfort by calculating the average jerk.

The Pre-Trained row in Table III shows the test performance of AutoBot in both regular and adversarial scenarios, as previously introduced in Section IV-C. We employed expert trajectories generated in safety-critical scenarios, performed data cleaning, and fine-tuned the planner. The fine-tuning results, presented in the "Fine-Tuned" row of Table III, show a slight increase in OR compared to the pre-trained planner, likely due to the incorporation of numerous safety-critical trajectories, resulting in a more aggressive driving style. The fine-tuned model achieved lower PCR, higher AV, and reduced jerk in both regular and adversarial scenarios. These outcomes demonstrate that the generated expert trajectories enhance the safety, efficiency, and comfort of autonomous driving planners, thereby validating STD's effectiveness in augmenting the data distribution.

Figure 7 displays the visualization of the pre-trained and fine-tuned planners in safety-critical scenarios. In the lane-change scenario, the pre-trained planner collides with another agent during the lane change, while the fine-tuned planner successfully avoids the collision through better trajectory planning. In the intersection scenario, the pre-trained planner fails to respond properly in the complex intersection environment, resulting in a collision between the ego agent and other agents. In contrast, the fine-tuned planner handles the intersection more effectively, ensuring the ego agent passes through safely. In the following scenario, the pre-trained planner fails to detect that agent 5 accelerates uncontrollably, leading to a rear-end collision with the ego agent. The fine-tuned planner, however, reacts quickly and accelerates in time to avoid the collision, allowing the ego agent to safely cruise in its lane (by the time agent 5 reaches the collision point, the ego agent has already passed through). These comparisons demonstrate that the fine-tuned planner significantly improves its ability to handle complex traffic situations. This further validates the effectiveness of STD in augmenting the data distribution.

E. Analyzing Safety-Critical Scenario

In this subsection, we conducted a comprehensive analysis of the safety-critical scenarios generated by STD. First, we filtered out scenarios where STD failed to generate collisions. Then, inspired by [69], to further categorize and analyze the types of collisions, we applied K-Means [70] clustering, setting the number of clusters to $k = 7$. Based on the clustering results, and following the classification of collision types from [64], we manually assigned semantic labels to the clusters. The examples of each collision type are displayed at the bottom of Figure 8. The green bars (left side) represent the proportion of each collision type among the total successfully generated safety-critical scenarios, while the orange bars (right side) represent the proportion of each collision type for which an expert solution was found to avoid the collision.

Specifically, we identified seven types of collisions: rear-end collisions (by adversary), head-on collisions, rear-end collisions (by ego), head-side collisions, T-bone collisions,

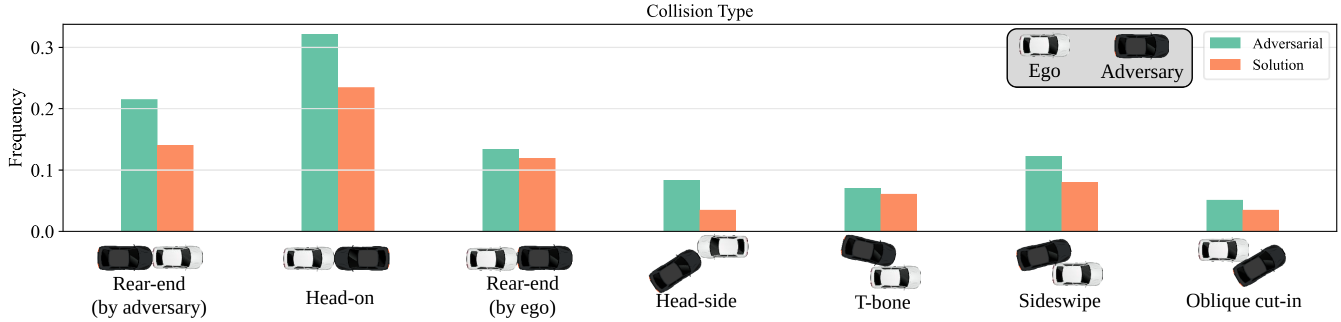


Fig. 8. Frequency and Solution Rate of Different Collision Types in Safety-Critical Scenarios

TABLE IV
EVALUATION OF DIV. AND LONG-TERM PERFORMANCE

Method	FR ↓	CR ↓	OR ↓	Cov ↑	TD ↑	RB ↓
SimNet	24.58	15.80	3.05	395.2	0.00	0.098
TrafficSim	27.08	20.39	3.42	861.0	4.28	0.119
BITS	14.82	8.62	**0.95**	1078.7	4.66	0.120
CTG++	14.35	9.17	1.45	**1518.9**	**7.13**	**0.069**
STRIVE	15.27	10.55	1.20	937.4	5.97	0.096
STD	**11.21**	**6.16**	1.71	1089.8	6.54	0.086
Dataset	17.09	15.95	0.50	539.3	0.00	0.000

sideswipe collisions, and oblique cut-in collisions. STD is capable of generating a diverse range of collision types, with head-on collisions being the most frequent at 29.67%, and oblique cut-in collisions being the least frequent at 4.75%. Rear-end collisions (by ego) had the highest rate of finding an expert solution, reaching 88.10%, while head-side collisions were the most challenging to resolve, with only 42.31% of cases finding a solution. In summary, the experimental results demonstrate that STD not only generates diverse collision types but also achieves high success rates in producing expert trajectories to resolve them across most collision categories.

F. Diversity and Long-Term Performance

In this subsection, we evaluate the diversity and long-term performance of scenarios generated by STD and baseline methods, extending the simulation timestep to 20 seconds to assess both aspects.

Metrics. **Diversity** is evaluated by coverage and trajectory diversity.

Coverage (Cov) evaluates how well the agents cover the scene. We estimate the trajectory distribution density using Gaussian Kernel Density Estimation (KDE) across all timesteps, focusing on the 2D spatial distribution. Coverage is defined as the number of grid points where the KDE estimate exceeds a threshold. In our experiments, the map is discretized with a 2m x 2m resolution, and we accumulate the results over 5 trials to calculate coverage density following [63].

Trajectory Diversity (TD) is calculated by measuring the Wasserstein distance between the density distributions of trajectories from different trials. Based on the previous study [45], we unfold all grid points with non-zero density, calculate the Euclidean distance between each pair, and generate a distance matrix. The density distributions are normalized to sum to 1, and the Wasserstein distance is calculated. For n trials of the same scene, the average Wasserstein distance between $n(n-1)/2$ pairs of density distributions is used as the diversity metric, expressed as:

$$\text{Diversity} = \frac{2}{n\left(n-1\right)} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \text{Wass}\left(\rho_i, \rho_j\right) \tag{30}$$

where $\text{Wass}(\cdot,\cdot)$ is the Wasserstein distance, and ρ_i is the density distribution of the i-th trial.

Table IV presents the quantitative results of long-term closed-loop simulations for generating regular scenarios. To account for environmental differences, we re-evaluated all baseline methods within a consistent setting. Additionally, we report results on a real-world dataset (Dataset), which exhibits noise as indicated by a non-zero failure rate, primarily due to agent collisions from inaccurate bounding box annotations. Notably, STD achieved a lower failure rate and generated more diverse scenarios compared to the real-world dataset. In our comparisons, STD outperformed SimNet and TrafficSim across all metrics. Although CTG++ slightly surpassed STD in diversity and realism, it fell short in terms of stability. BITS showed slight advantages in OR and Cov metrics, while STRIVE performed marginally better in OR but lagged behind STD in other metrics. Importantly, STD achieved the best results across all baseline methods in the FR and CR metrics, highlighting its superior stability, particularly in preventing agent collisions during long-term scenario generation.

G. Ablation Study

Impact of Hybrid Guidance. We investigated the contribution of each component in the hybrid guidance function of the safety-critical scenario generation module by evaluating the following variations: Specifically, we considered the following variations:

- STD-w/o-I. We removed the initialization guidance.
- STD-w/o-A. We removed our adversarial guidance and adopted an adversarial guidance similar to STRIVE.
- STD-w/o-C. We removed the collision guidance that avoids collisions with other vehicles.
- STD-w/o-E. We removed the environmental guidance that avoids collisions with the environment.

Through the ablation study on hybrid guidance in Table V, we can draw the following conclusions: The adversarial guidance is crucial for generating safety-critical scenarios

TABLE V
ABLATION OF HYBRID GUIDANCE

Method	GCR ↓	RB ↓	AOR ↓	OFR ↓	OCR ↓
STD	**73.00**	0.060	**0.01**	**5.18**	**4.65**
STD-w/o-I	72.00	0.080	0.02	5.67	5.20
STD-w/o-A	70.00	0.071	0.04	6.30	5.70
STD-w/o-C	72.00	**0.051**	0.07	7.74	7.27
STD-w/o-E	72.00	0.077	1.82	9.16	5.83

TABLE VI
ABLATION OF STD COMPONENTS

Method	FR ↓	CR ↓	OR ↓	Cov ↑	TD ↑	RB ↓
STD	**11.21**	**6.16**	1.71	1089.8	6.54	**0.086**
STD-w/o-D	12.44	7.90	**1.65**	1021.2	6.04	0.093
STD-w/o-S	14.61	9.64	1.77	**1639.6**	**7.03**	0.103

with a high collision rate, as it increases the likelihood of collisions and reduces other failure rates and unrelated collisions. The environmental guidance plays an important role in ensuring the realism of the scenarios, reducing off-road incidents, and lowering the failure rate. The initialization guidance has a significant impact on the initial setup of the model and the reasonableness of agent behavior, particularly in preventing off-road incidents and reducing other collision rates. Therefore, the complete hybrid guidance performs best across multiple metrics, particularly in maintaining realism and generating effective safety-critical scenarios with a well-balanced performance.

Impact of Components. We further assessed the role of specific components within STD, namely the Social Transformer and Decoder, by evaluating the following variations:

- STD-w/o-D: STD without the Decoder component.
- STD-w/o-S: STD without the Social Transformer component.

As shown in the long-term scenario generation ablation study in Table VI, the Social Transformer is crucial for significantly reducing the FR, CR, and maintaining traffic realism. Removing the Decoder has a smaller impact on several metrics but still contributes to both Cov and diversity. Overall, the full STD model demonstrates the best balance across all metrics, generating stable, realistic, and diverse scenarios.

V. RELATED WORK

Methods for obtaining safety-critical scenarios can be divided into two categories: real-world data collection and simulation-based generation.

The real-world data collection approach samples and extracts safety-critical scenarios from daily driving data. Works like [16]–[20] use historical analysis and clustering to identify sparse, low-frequency scenarios. However, safety-critical scenarios are rare in daily driving data, making extraction difficult and time-consuming. Moreover, the limited diversity of real-world scenes means sampled data may not cover all extreme cases, leaving gaps in autonomous system training [71].

Another approach is to generate safety-critical scenarios in a simulated environment. Using simulation tools like CARLA [72], it is possible to manually design the paths of autonomous vehicles to create such scenarios [73]–[75]. However, when large numbers of complex scenarios need to be generated, manually designing paths becomes both time-consuming and labor-intensive, making it difficult to meet the growing demand for diverse scenarios in autonomous driving systems [76].

Consequently, recent research has increasingly focused on automating the generation of safety-critical scenarios to efficiently generate diverse scenes. [21], [22] generate safety-critical scenarios by initializing opponents' states. [23] applies adversarial optimization in each step of a Diffusion model with a U-Net [77] architecture. [24] uses kinematic gradients to modify vehicle trajectories, improving imitation learning robustness. [25] leverages LLMs to generate safety-critical scenarios by converting language into control codes. [26] combines traffic priors and CVAE [78] to optimize adversarial trajectories for realistic scenarios. [27] transforms historical trajectory replays into safety-critical scenarios using resampling. [58] uses Bayesian optimization to disrupt vehicle paths by modifying trajectories. [29] models human driving strategies with GAIL [79] and designs reward functions to generate safety-critical scenarios.

However, these methods have several limitations. [21]–[25] overlook dynamic factors in complex traffic environments, particularly in multi-vehicle interaction scenarios. Adversarial vehicles may prematurely collide with non-autonomous vehicles in these environments, leading to unrealistic and implausible generated scenarios. [26], [27] heavily rely on original trajectories, which limits their diversity when generating safety-critical scenarios. Furthermore, without these original trajectories, they are unable to produce meaningful data. Lastly, [21], [28]–[30] do not adequately assess the actual impact of safety-critical scenarios on the performance of autonomous driving planners, which is a primary concern for users. These methods lack end-to-end testing on autonomous driving planners. Additionally, they fail to validate the contribution of the generated scenarios to the improvement of planner performance. Advancements in generative models have made controllable traffic simulation possible. CTG [37] guides Diffusion to generate controllable traffic simulations using STL [80], [81] rule analysis loss functions [37] or LLMs. While CTG can generate goal-oriented trajectories, it struggles with efficiently handling specific safety-critical scenarios.

VI. CONCLUSION

In this work, we propose Scenario Transformer Diffusion (STD), a novel method for generating safety-critical scenarios. STD is developed based on a query-centric multi-agent traffic generation model, capable of generating stable, realistic, and diverse safety-critical scenarios, along with expert trajectories. Extensive experimental results demonstrate that STD surpasses current state-of-the-art methods in terms of stability, realism, and diversity. Additionally, STD not only effectively tests the performance of autonomous driving planners but also enhances the data distribution through its generated expert trajectories.

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